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Abstract

TripMate Macau is a proposed intelligent travel planning system for independent travelers visiting Macau. It will turn trip information and selected attractions into a time-aware and cost-aware daily itinerary. Preference matching will help users discover suitable points of interest (POIs), while separate Favorite and Add actions will distinguish saved places from current-trip targets. Users will be able to adjust visit times or transport modes, with downstream times and availability checks recalculated automatically. A geographic Macau map will follow the itinerary order. Companion information entered during setup will support Travel Bill records and settlement in CNY, rather than affect route planning. A lightweight Itinerary Plaza will support publishing and browsing plans and bookmarking individual POIs. Four members will own user-facing functional areas and develop them through parallel work and regular integration. Evaluation will cover approximately 30 Macau POIs, constraint-processing cases, expense calculations and the complete workflow. Existing prototypes provide a starting point; the final design describes proposed behavior rather than a completed integrated system.

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Chapter 1 Introduction and Problem Diagnosis

1.1 Background

Independent travel gives visitors freedom to choose where to go and how long to stay. However, this freedom also places the planning work on the traveler. A Macau visit may include attractions across Macau Peninsula, Taipa, Cotai and Coloane. A useful plan must connect these choices within the time and money available.

This proposal diagnoses the effort required to turn travel information into an executable itinerary, proposes a treatment, and sets out a practical plan of work.

1.2 Current Travel Planning Practice

A traveler may collect suggestions from travel websites, map services and social-media guides. After choosing places of interest, the traveler checks their locations, opening hours, entry conditions and prices. The traveler then estimates visiting duration and transport needs, arranges a visiting order and fits the visits into each day.

These decisions are connected. Changing one attraction can affect the arrival time at every later stop. Group travelers also need to record who paid for meals, tickets or transport and decide how to divide the costs. Current planning therefore involves both collecting information and repeatedly reconciling it.

1.3 Problems in Current Practice

Information fragmentation makes it difficult to compare attractions consistently. A guide may suggest a place without stating its last-entry time, while a map may show its location without explaining how long a visit usually takes. Missing information can remain unnoticed until the traveler tries to build a timetable.

Geographical order alone is insufficient. For example, a nearby attraction may close before a later stop can be completed. A traveler who chooses attractions in preference order may also cross between the peninsula and the islands repeatedly. Reducing distance, meeting opening hours and staying within a budget require different checks.

Manual revision adds further effort. Removing a stop or changing the starting time may require recalculating the whole day. The specific problem addressed here is the manual effort of transforming fragmented travel information into a feasible, time-aware and cost-aware daily itinerary.

1.4 Target Users

The main users are independent travelers who are unfamiliar with Macau and have limited time to prepare. First-time visitors need help relating attractions to areas and visiting times. Small groups need a common plan and a clear record of shared expenses. The first version supports these users within Macau rather than attempting worldwide coverage.

1.5 Project Objective

The objective is to reduce the effort of finding suitable attractions and turning them into a feasible daily plan. The system will process preferences, arrange selected POIs, and recalculate schedules after visit-time or transport changes. Users will retain control over their choices, see unresolved constraints, and manage group expenses through companion-linked bills and CNY settlement. Chapter 7 defines how these outcomes will be assessed.

Chapter 2 Proposed Treatment

2.1 Solution Overview

TripMate Macau will combine trip dates, a departure point within Macau, daily available time, preferences and budget with structured POI information. Users may choose History & Culture, Food, Shopping or Nature. The system will process these preferences and display understandable values such as 90% Match on POI cards. These values describe preference suitability, not route feasibility or a tested accuracy rate; users will not need to interpret a raw scoring formula.

A heart action will save a POI to Location Favorites, while a plus action will add it as a target for the current trip. The system will arrange only the current-trip targets into a schedule and check known opening windows and entry restrictions. If not all targets fit, it will identify unresolved choices rather than present an impossible plan as feasible. Users may also discover individual POIs through published itineraries in Itinerary Plaza, without copying another user's whole itinerary.

2.2 Typical User Scenario

A first-time visitor opens Itinerary Planning and enters travel dates, a Macau departure point such as Border Gate Port, daily start and end times, budget and preferences. The visitor manually adds companions for later bills. On the POI Selection page, the visitor compares preference Match values, hearts a POI for a later trip and uses plus to add another POI to the current trip. Favorites and trip targets remain separate. The visitor reviews the selected targets before confirming generation.

After generation, the visitor reviews My Macau Itinerary and its map. Changing a connection from Bus to Taxi will update the estimated travel time, recalculate later arrival and departure times, and recheck opening hours, last entry and the daily end time. The visitor may also change a POI's start or end time in 15-minute intervals, or delete it. During travel, bills will use the companions entered during setup; detailed CNY transfers will appear only after opening Settlement / Split Bill.

2.3 Expected Benefits

Preference matching will reduce manual comparison of POIs, while Location Favorites will preserve places independently of a particular trip. Structured information and automatic scheduling will reduce repeated searches and timetable calculations. Transport changes will trigger downstream recalculation and updated warnings, saving users from checking every later stop manually. A geographic map that follows the itinerary will help users understand the route.

The lightweight Plaza will support discovery through other travelers' published plans without treating a complete itinerary as automatically suitable for a new trip. Reusing companion data for bills and normalizing settlement to CNY will reduce repeated entry and arithmetic. These are expected benefits to be evaluated, not claims of proven savings or guaranteed optimal routes. Evaluation will record task time, corrections and user feedback.

2.4 Project Scope

The first version will cover Macau through the four main functional areas and supporting capabilities in Chapter 3. These include Match-based discovery, Location Favorites, a lightweight Itinerary Plaza, automatic planning and recalculation, a geographic map, and companion-linked Travel Bill and settlement. Planning will use a controlled initial POI dataset, verified availability where obtainable, and clearly labeled travel-time and cost estimates. Other cities may be added later; city expansion is outside initial acceptance.

Hotel and flight booking and payment execution are excluded. A full social networking system is outside the MVP scope. However, TripMate includes a lightweight Itinerary Plaza for publishing and browsing itineraries and discovering individual POIs. Users cannot copy an entire published itinerary into their own trip. Xiaohongshu import, weather-aware automatic replanning and live transit information remain future work. No live traffic or turn-by-turn navigation service is promised.

Chapter 3 Functional Features

The four main workflow stages remain enter trip information, select target POIs, generate an itinerary, and adjust and save the plan. Each stage has one functional product owner. Location Favorites, Itinerary Plaza, the geographic map and Travel Bill support this workflow without becoming technical ownership departments. The following behavior reflects the proposed final product design.

3.1 Trip Information, Preferences and Companion Data

Users will create, save, reopen and update a Macau trip. Inputs will include dates, daily available start and end times, departure location, budget and preferences such as History & Culture, Food, Shopping and Nature. Departure means a point within Macau: users may select Border Gate Port, Hengqin Port, Hong Kong-Zhuhai-Macao Bridge Port, Macau International Airport, Taipa Passenger Terminal or Outer Harbour Passenger Terminal, or manually enter another Macau location. The system will validate incomplete inputs and inconsistent time ranges. Member A owns this outcome.

Users will manually add companions during trip setup. These records will populate payer and participant choices in Travel Bill, expense splitting and settlement. Companions will not participate in automatic itinerary planning: Trip Input → Companion Data → Travel Bill is a separate data path from the trip constraints used to generate a route.

3.2 POI Discovery, Match, Favorites and Selection

Within Itinerary Planning, the POI Selection page will let users browse, search and filter Macau POIs and review location, opening hours, applicable last-entry information, recommended visiting duration, price and availability. Preference processing will produce user-facing Match values, such as 90%, 85% or 78%, to support comparison. Within each category, POIs will be ranked in descending order according to their calculated Match values, prioritizing attractions that better fit the user's preferences. The scoring calculation will be performed by the system without exposing its raw formula. Match does not confirm that a POI fits a particular day.

Verified, estimated and missing availability information will remain distinguishable. Member B owns this outcome.

Favorite and Add are separate actions. The heart will save a POI to Location Favorites for later review; the plus will add it to the current trip's target attractions. Favoriting a place will not schedule it, and adding it to a trip will not silently change the saved favorites collection. Location Favorites will be accessible from the homepage.

The bottom-navigation Discover page will provide broader content discovery, including published itineraries and attraction details. It is separate from the trip-specific POI Selection page. Itinerary Plaza will let users publish saved itineraries, browse other users' published plans and view their maps. A user may heart or bookmark an individual POI from a published itinerary. The whole itinerary cannot be copied directly into the current trip. Publishing and browsing are limited sharing capabilities, not a full social network.

3.3 Automatic Itinerary Generation and Constraint Processing

The system will organize current-trip target POIs into daily schedules and a reasonable visiting sequence. It will consider departure point, available time, location, opening windows, last entry, recommended visiting duration, estimated travel time and cost, preferences and budget. The complete visit must fit known closing and daily end times. Location Favorites will not enter the planner unless the user adds them to the current trip, and companion names will not affect planning. Member C owns generation and constraint processing.

If the current selection cannot fit, the system will identify unscheduled places or unresolved constraints and explain which inputs need adjustment. Unknown opening hours and costs will remain visible as uncertainties. The proposal does not promise a mathematically optimal route or verified live transport information.

3.4 Itinerary Management, Map and Recalculation

Users will review My Macau Itinerary and edit individual POIs or transport segments before saving the revised plan. Clicking an attraction will open three actions:

Modify Time, Record Expense, and Delete POI. Under Modify Time, users can adjust the POI's start and end times in 15-minute intervals, such as 10:00, 10:15, 10:30 and 10:45. These are explicit start and end times; recommended duration remains planning information. An invalid time range will require correction. Share Itinerary, Replan and Map will be available as bottom actions within the itinerary page. Member D owns itinerary management and the coordinated update of its views.

Each connection between POIs will show transportation mode, estimated travel time and map/navigation access. Clicking the mode will allow the user to change it. When a user changes Bus to Taxi, the system will obtain the revised travel estimate, recalculate downstream itinerary times, recheck opening-hour, last-entry and daily-end constraints, and update conflicts or warnings. Changes to POI times or deletion will also trigger relevant checks. Replan will request a newly generated schedule using the current trip inputs and targets. If an estimate is unavailable, the system will show uncertainty rather than claim the schedule is verified.

The itinerary's Map button will open an actual Macau geographic base map with numbered POI markers and route connections. If the itinerary is 1→2→3→4→5, the map must use that same order. Map/navigation access will help users inspect the connection; it does not imply live traffic or live turn-by-turn guidance. Bottom navigation will contain four tabs: Home, Discover, Map and Mine. My Macau Itinerary is a functional page reached from the homepage, after generation, or through Mine; it is not a bottom-navigation tab. The daily itinerary may display its date, weather condition and temperature for reference, but weather-aware automatic replanning is outside the MVP scope.

Travel Bill will accept a location, payer, amount, original currency (MOP, HKD or CNY) and participating companions. After participants are selected, the system will automatically calculate the number of participants and the equal per-person share for that expense. Its main page will show total expenditure, expense records, Add New Bill and a Settlement / Split Bill entry point. It will not display final who-pays-whom transfers. Opening Settlement / Split Bill will show each person's actual payment, required share and final transfer relationships normalized to CNY/RMB. Original amounts and applied conversion rates will remain traceable; no live exchange-rate service is assumed.

The proposed project will allow users to:

• Manage Macau trip inputs, preferences and companions, validating dates and daily times.

• Browse, search and filter POIs, compare preference Match values and add targets to the current trip.

• View POI details and save individual places separately in Location Favorites.

• Generate a daily itinerary automatically and report unresolved time, availability or budget constraints.

• Edit POI times or transport modes, recalculate later times, and save or review the updated plan.

• View numbered POIs and route connections on a geographic Macau map in itinerary order.

• Record bills in MOP, HKD or CNY and open Settlement / Split Bill to review CNY shares and transfers.

• Publish and browse Plaza itineraries and bookmark individual POIs without copying an entire itinerary.

3.5 Proposed User Interface and Workflow

Figures 3-1 to 3-3 show selected screens from the current front-end design. Figure 3-1 connects the homepage, trip inputs and trip-specific POI Selection. Figure 3-2 shows itinerary review, individual POI editing and a geographic Macau map. Figure 3-3 illustrates the supporting Plaza, Travel Bill and Mine pages. Chinese labels and sample values are retained. These screens represent the proposed interface and interactions; they do not establish that the complete integrated system or displayed sample results are implemented. Map markers and route connections must follow the generated itinerary order.

(a) Home



(b) Trip Information



Figure 3-1 Trip information and POI Selection interface design.

(c) POI Selection

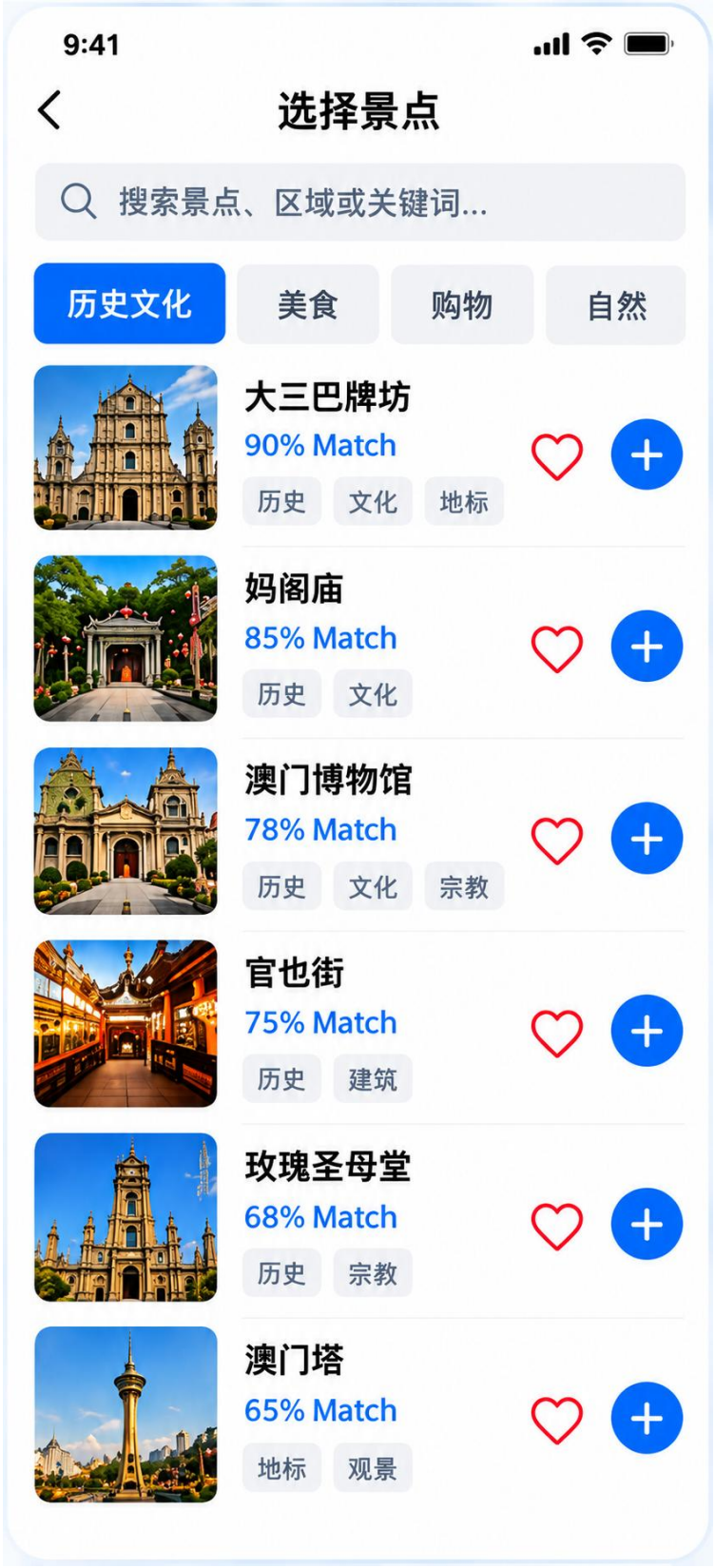


Figure 3-1 (continued).

(a) My Macau Itinerary

(b) Edit individual POI



Figure 3-2 Itinerary editing and geographic map interface design.

(c) Itinerary Map



Figure 3-2 (continued).

(a) Itinerary Plaza

(b) Travel Bill

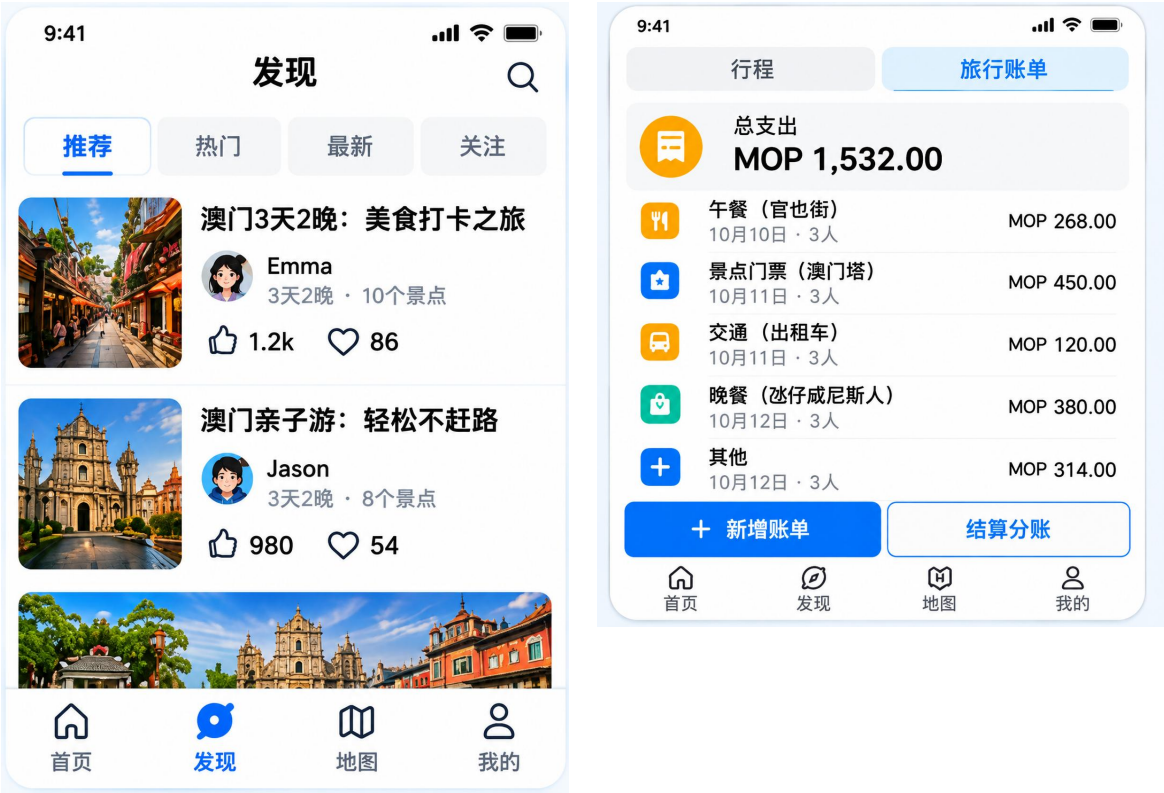


Figure 3-3 Itinerary Plaza, Travel Bill and Mine interface design.

(c) Mine



Figure 3-3 (continued).

Chapter 4 System Usage Flow

4.1 Overall User Flow

The homepage will provide Itinerary Planning, Location Favorites, My Macau Itinerary and Itinerary Plaza. Bottom navigation will contain four tabs: Home, Discover, Map and Mine. Discover is a broader content entry, while POI Selection belongs to Itinerary Planning. The main planning path is Itinerary Planning → Enter Trip Information → POI Selection → Confirm and Generate Itinerary → Review / Edit Itinerary, summarized in Figure 4-1. My Macau Itinerary can also be opened from the homepage or Mine. Favorites are a separate saved collection; a place becomes a planning target only through Add. Users can open a published Plaza itinerary and its map, then bookmark individual POIs.

Step 1 Enter trip information

↓

Step 2 Select POIs

↓

Step 3 Generate itinerary

↓

Step 4 Adjust and save itinerary

Figure 4-1 Overall traveler workflow.

4.2 Automatic Travel Planning Flow

The system will validate trip information, obtain current-trip targets and check their date-specific availability. It will combine visiting duration with estimated transport time and cost to construct a schedule. Preference Match supports discovery, while constraint processing determines whether selected visits fit. Companion data will be excluded from planning and reused by Travel Bill instead.

The generated result will supply ordered stops, start and end times, travel estimates and warnings to both the itinerary and map. Editing a transport mode will change the travel estimate, propagate the resulting time change to downstream visits, and rerun availability and daily-end checks. Editing or deleting a POI will similarly

update the affected schedule. Warnings must reflect the revised plan rather than remain tied to the original result.

4.3 Example Macau Travel Scenario

Consider an illustrative one-day trip from Border Gate Port. The traveler chooses History & Culture and Food, compares Match values on the POI Selection page, and adds selected attractions to the current trip. A different place is hearted for Location Favorites only. A POI discovered on a published Plaza map may also be bookmarked individually; the published itinerary cannot be copied wholesale.

The planner will use the chosen date and verified POI records to decide which sequence fits. If the traveler changes a Bus connection to Taxi, its revised estimate will shift downstream times and update opening-hour, last-entry and daily-end warnings. The traveler may set a POI's start to 10:15 and end to 11:00, or remove it and request an updated plan. The geographic Macau map must show the resulting stop order. These example times are input choices, not verified attraction opening hours.

For a controlled settlement example, a traveler records a MOP 240 meal paid by Alice and shared by Alice, Bob, Cathy and David, all entered as companions. Using a test-only rate of 1 MOP = CNY 0.90, the normalized total is CNY 216 and each share is CNY 54. Travel Bill shows the expenditure summary and record. Only Settlement / Split Bill shows Alice's actual payment of CNY 216, the others' zero payments, each required share, and the three transfers of CNY 54 to Alice. This rate is a calculation fixture, not a current exchange-rate claim.

Chapter 5 Plan of Work

5.1 Development Strategy

The team will develop the proposed final design through four functional ownership areas. Member A owns trip information, preferences, companion data and supporting bills; Member B owns POI discovery, Match, favorites, selection and Plaza discovery; Member C owns automatic generation and constraint processing; Member D owns itinerary management, map and recalculation. Each owner will design, implement, test and document the relevant outcome. The current interface sketches guide user-facing behavior; earlier prototypes provide preliminary work to reuse where suitable, rather than evidence that the expanded functions are complete.

The team will agree on target-POI and favorite identifiers, preference inputs, planning output, transport estimates, warning updates, companion records and bill currencies. Integration will follow the current four-tab navigation and keep Discover separate from trip-specific POI Selection. The geographic map and separated billing/settlement flow require integration and validation. No external data or live service is assumed to be available merely because a screen exists.

5.2 Task Dependencies and Parallel Development

After agreeing on shared examples, all four functional areas can progress in parallel. A can develop validated trip inputs, companion management and bill records. B can develop Match display, favorites, current-trip selection and Plaza browsing against sample records. C can develop planning against a small valid dataset, without waiting for the full initial POI dataset. D can develop POI editing, the map and recalculation interactions against sample planning output. Technical work remains an activity within each feature.

Integration will first connect trip inputs and current-trip POIs to generation, then feed the resulting sequence to My Macau Itinerary and the map. Recalculation depends on agreed transport estimates and C's constraint-checking behavior. Separately, A's companions will supply payer and participant choices for Travel Bill. Settlement requires defined conversion rates, rounding and balances; it does not depend on route

planning. Plaza publishing must use saved itinerary content, while bookmarking must update individual POIs rather than copy the itinerary.

5.3 Development Milestones

Table 5-1 gives a provisional six-week estimate, measured from the team's agreed development start. The plan reserves four members for parallel feature work and a final refinement period.

Table 5-1 Provisional feature development milestones.

Period	Functional work and owners	Reviewable result
Week 1	A–D agree preferences, targets/favorites, POI availability, companion/bill records, transport estimates and warnings.	Shared examples and acceptance cases for the final design.
Week 2	A: validated inputs and companions. B: Match, Favorite/Add and discovery. C: basic planning. D: itinerary editing and map prototype.	Parallel feature increments using agreed sample records.
Week 3	A–D connect inputs → target POIs → planning → itinerary/map; connect companions to bill inputs.	Actual planning output and independent companion-to-bill workflow.
Weeks 4–5	A: currency bills and CNY settlement. B: approximately 30 representative POIs, favorites and Plaza browsing. C: constraints. D: downstream recalculation, geographic map and Plaza publishing.	Integrated final-design capabilities and updated warnings.
Week 6	Each owner tests and documents their features; all members check integration, Match, Plaza and settlement cases.	Demonstration, verified calculations and evaluation results.

Development will use existing computers, browsers, prototypes and freely available tools. No paid data or hosting purchase is currently planned. Any service charge discovered during source verification will be reviewed before adoption. A detailed monetary budget and individual weekly effort have not yet been agreed, so no unsupported total cost or working-hour commitment is stated.

5.4 Integration Strategy

Each integration checkpoint will demonstrate a user task across feature owners: preference matching and selection; itinerary generation and geographic map order; transport changes and revised warnings; or companion-linked billing and CNY settlement. A separate check will publish and browse a saved Plaza itinerary and bookmark one POI. The team will review changes before combining them and keep a runnable version, following the team's agreed collaboration approach.

Each weekly checkpoint will demonstrate a customer task and record defects, decisions and the next priority. Integration is a recurring activity involving all four feature owners. It will not be postponed until every feature is individually finished.

5.5 Risks and Adjustments

Incorrect or outdated POI information could produce an impractical plan. The team will record sources and verification status, check demonstration POIs and flag uncertain data. Match values must be explained as preference suitability, not verified availability or an accuracy guarantee. Missing hours and prices will not be filled with invented values.

Transport estimates, map availability and currency conversion may be uncertain. The system will label estimates and use controlled fixtures for acceptance testing without claiming live traffic or current rates. Bill records will retain their original amount and currency, with the applied conversion basis visible in settlement. If conversion is unavailable, unresolved amounts must be flagged before final settlement. Recalculation tests will cover changes that introduce or remove downstream conflicts.

The expanded design adds integration work for favorites, Plaza, companion reuse and settlement. The team will build a small complete path first, then expand the dataset and supporting cases. Weekly reviews will rebalance implementation assistance while keeping feature accountability clear. Full social networking, live transit, weather-aware automatic replanning and text import will remain outside the initial scope rather than displace the agreed final-design capabilities.

Chapter 6 Team Profile and Product Ownership

6.1 Team Profile

The proposed team has four members: ( ), ( ), ( ), and ( ). The team name is Vibecoding and the group number is D2-Team2. The project website or repository address is https://github.com/lzq2005/CS250-SE.

The team members have complementary strengths. Member A, , has strengths in programming, project organization and documentation. Member B, , has strengths in user-interface design, presentation and functional testing. Member C, , has strengths in programming, algorithmic logic and testing. Member D, , has strengths in interface design, integration and documentation. These strengths indicate areas in which each member may contribute effectively, but all members will participate in requirements analysis, design, implementation, testing and documentation according to project needs.

Four members allow the traveler-facing workflow and the information-and-planning workflow to advance together while retaining time for review and integration. A smaller team would need to combine these responsibilities and reduce the work performed in parallel. All members are expected to contribute to requirements, design, implementation, testing and documentation according to project needs.

6.2 Team Leader

The team leader is . The leader will facilitate meetings, track progress and unresolved decisions, and coordinate integration checkpoints. This role does not replace the leader's customer-visible feature responsibility or imply that one person performs all documentation and testing. will submit one report on Moodle on behalf of D2-Team2.

6.3 Individual Product Ownership

Product ownership identifies the member accountable for a customer-visible functional outcome, its progress and its acceptance. Technical work is an activity

needed to deliver that outcome, not the final division of the product. Table 6-1 assigns the four workflow stages to Members A to D.

Table 6-1 Functional product ownership.

Owner	Product ownership	Customer-visible result
A	Trip Information, Preferences & Companion Data	Validated trip inputs and companions; Travel Bill and CNY settlement.
B	POI Discovery, Match, Favorites & Selection	Preference-based POI discovery, independent Favorite/Add and Plaza browsing.
C	Automatic Itinerary Generation & Constraint Processing	Feasible daily schedules and reusable checks with transparent warnings.
D	Itinerary Management, Map & Recalculation	Edited and saved plans, consistent geographic map, downstream updates and Plaza publishing.

Member A will own Trip Information, Preferences & Companion Data, including validated dates, Macau departure point, daily times, preferences and manually added companions. A will also own Travel Bill and Settlement / Split Bill: original-currency inputs, payer and participant reuse, summary pages, CNY conversion, shares and final transfers. Input screens, storage, currency calculations, testing and documentation are activities beneath these customer-visible outcomes.

Member B will own POI Discovery, Match, Favorites & Selection. B will deliver browsing, search, filtering, POI details, understandable Match values, separate heart and plus actions, and Location Favorites. B will also own Plaza browsing, published-itinerary detail and map access, and individual-POI bookmarking. Preparing POI records, implementing preference matching, testing state separation and documenting the feature are supporting activities.

Member C will own Automatic Itinerary Generation & Constraint Processing. C will deliver daily order, scheduled visits, travel estimates and transparent feasibility warnings, excluding companion data from planning. C will also provide the shared checks for edited plans and work with D on downstream updates. Planning logic,

input/output handling, boundary-case tests and documentation remain activities within this ownership.

Member D will own Itinerary Management, Map & Recalculation. D will deliver explicit POI start/end editing, applicable expense access, deletion, transport-mode changes, Replan, saved plans and the Macau geographic map. D will coordinate time propagation and rerun C's checks so itinerary, map and warnings stay consistent. D will also own publishing a saved itinerary to Plaza through Share Itinerary, coordinating with B's browsing feature. Interface work, data updates, integration tests and documentation support these outcomes.

Supporting work is distributed to keep the workload reasonable: bills and settlement are linked to A's companions; Plaza discovery extends B's POI discovery; reusable constraint checks remain with C; and map, recalculation and publishing extend D's itinerary management. The team will review effort weekly and share implementation, testing and documentation work without replacing functional ownership with technical departments.

6.4 Pair Arrangement

Pair 1 consists of A and B and contributes trip inputs, preferences, companion reuse, discovery, Match, favorites and selection. Its supporting work includes Travel Bill, CNY settlement and Plaza discovery. Its quality focus is clear user actions, consistent data reuse and correct expense calculations.

Pair 2 consists of C and D and contributes generation, constraint processing, itinerary editing, downstream recalculation and map consistency. D's Plaza publishing will integrate with B's discovery work. The pair's quality focus is transparent warnings and consistent schedule updates rather than merely changing displayed labels.

All members will participate in requirements, design, implementation, testing and documentation. The pairs support collaboration rather than separate technical departments. Shared examples and regular joint demonstrations will connect all four stages of the traveler's workflow.

6.5 Contributions to This Report

Table 6-2 Contributions to This Report

Member	Student ID	Contribution to This Report
		Coordinated the proposal, integrated the final document, reviewed the project requirements and functional design, and completed the final formatting and submission preparation.
		Contributed to the user-interface design and prepared/reviewed the UI prototype materials and related functional descriptions.
		Contributed to the automatic itinerary generation and constraint-processing descriptions and reviewed the corresponding evaluation criteria.
		Contributed to itinerary management, map/recalculation descriptions and supporting documentation/review.

Chapter 7 Evaluation and Success Criteria

7.1 Functional Evaluation

The acceptance dataset will contain approximately 30 Macau POIs. Each record must have a unique identifier, name, location and category, with opening information, last entry where applicable, recommended duration and price either supplied or explicitly marked as missing or uncertain. Verification status and source records will be checked before the final demonstration.

Functional checks will cover the four ownership areas and supporting capabilities. Users must be able to enter a Macau departure point, preferences and companions; compare Match values on POI Selection; heart a POI without adding it to the trip; add or remove current-trip targets without losing unrelated favorites; generate a plan; edit POI times and transport; save and reopen it; and see the same route sequence on the geographic map. Navigation checks will verify Home, Discover, Map and Mine as the four bottom tabs, distinguish Discover from POI Selection, and open My Macau Itinerary from its homepage and Mine entries. The workflow must use actual planning output rather than a fixed mock itinerary.

Bill tests will cover original amounts in MOP, HKD and CNY, companion-based payer and participant selection, subsets of participants, multiple payers and deletion. Tests will verify participant counts and equal per-person shares, including subsets of participants and rounding where necessary. Controlled conversion rates will be compared with hand calculations, including the CNY 216 meal example in Section 4.3. Original currencies must be retained; normalized totals and rounded shares must agree, and settlement balances must sum to zero. Travel Bill must show summaries and records without final who-pays-whom transfers. Only Settlement / Split Bill may show actual payments, required shares and final CNY transfers.

7.2 Planning Feasibility Evaluation

Controlled cases will cover closed dates, arrival after last entry, insufficient visit time, low budget, missing availability, edited POI start/end times and changed transport. A Bus-to-Taxi test will verify the revised travel estimate, propagation to every affected

downstream stop, and refreshed opening-hour, last-entry and daily-end checks. Cases must demonstrate both introducing and resolving a conflict. No plan with a known violation may pass as feasible.

If targets cannot fit, the result must identify unscheduled places or unresolved constraints. Unknown information must produce a warning. Admission and transport estimates must be summarized, and an over-budget plan must not be presented as meeting the budget. POI start and end times must be selectable in 15-minute intervals, and end-before-start input must be rejected. Changing only companion names must leave an otherwise identical planning request unchanged while updating bill-selection choices.

The geographic map must show the same POI identifiers, numbered order and route connections as the itinerary, including after POI time edits, deletion, transport changes and replanning. Preference tests will verify that changing categories updates the displayed Match results in accordance with agreed scoring cases and that POIs within each category appear in the expected descending Match order, without labeling those values as route feasibility. Favorite and Add states must remain independent. Plaza checks will publish and browse a saved itinerary, open its map and bookmark an individual POI; no whole-itinerary copy action may be offered. Weather information may be displayed for reference, but changing the displayed weather must not trigger weather-aware automatic replanning in the MVP.

7.3 User Interaction Evaluation

A small group of classmates will be invited to evaluate the main workflow: create a Macau trip, add companions, compare Match values on POI Selection, distinguish Favorite from Add, generate a plan, edit a POI time, change a transport mode, inspect the map and enter a bill. They will then open Settlement / Split Bill to find the final CNY result. A short Discover / Plaza task will ask them to bookmark one POI from a published itinerary. The evaluation will examine whether most participants can complete the main workflow without step-by-step assistance. No participants or results have yet been collected.

Participants will also complete a comparable manual planning task using the same POI information. The team will compare completion time, errors, corrections and feedback, using equivalent selections to reduce repetition effects. This small evaluation will guide refinement; no improvement is claimed before testing.

Chapter 8 Conclusion

TripMate Macau will reduce the work of turning travel preferences and fragmented POI information into an executable itinerary. Match values, separate favorites and trip targets, and a lightweight Plaza will support discovery. Automatic generation and downstream recalculation will keep visit times, availability warnings and the geographic map consistent as users adjust the plan. Companion-linked bills and CNY settlement will support group expenses.

The first version will remain Macau-focused and city-expandable. Existing prototypes provide a starting point, while the final design still requires integrated implementation and evaluation. Functional product ownership, parallel development and regular integration will guide this work. Success will be assessed through the complete user workflow, transparent constraint processing, independent Favorite/Add actions and correct billing and settlement, without claims of live services or completed performance results.